

Credit risk analysis - project guide

A simple explanation of the interactive credit-review dashboard and the machine-learning risk pipeline.

Dashboard

Review borrower strength, financial ratios, downside resilience, and credit conditions.

Machine learning

Train and use a calibrated model that estimates borrower risk from structured data.

What this project answers

Can a borrower repay debt? What makes them risky? What changes if revenue falls? CreditLens presents those answers in a clear review workflow.

GitHub repository: github.com/jayantjain05/CREDIT_RISK

1. The dashboard: a borrower review workspace

The web interface acts like a simplified internal tool for a bank credit analyst. It uses a sample business, Apex Components Private Limited, to show how a credit review can be organized.

Feature	Why it matters
Credit score	A quick 0-100 summary of financial health. A higher score indicates lower apparent risk.
Probability of default	An estimated chance that a borrower may become high risk or fail to repay.
DSCR	Shows whether cash available for debt payments is sufficient to cover scheduled debt obligations.
Risk grade	A simple rating such as A-, BBB, or BB+ to describe overall credit quality.
Financial trends	Revenue and EBITDA margin show whether the underlying business is strengthening or weakening.
Credit memo	A concise recommendation containing the proposed decision, risks, and approval conditions.

Key idea

The dashboard is a decision-support tool. It helps an analyst understand the borrower; it does not automatically approve or reject a loan.

2. Financial health and stress testing

CreditLens combines basic financial metrics with a downside scenario. The goal is to understand both the current position and how resilient the borrower may be during a business contraction.

Important measures

- **DSCR:** cash available for debt payments divided by total debt payments. A DSCR of 1.62x means approximately Rs 1.62 is available for every Rs 1.00 owed.
- **Net debt / EBITDA:** compares debt with operating earnings. Lower leverage is generally preferable.
- **Interest coverage:** measures the ability to pay interest from operating earnings. Higher coverage is generally preferable.
- **Current ratio:** compares short-term assets with short-term liabilities and provides a liquidity signal.

Stress-test flow

Revenue decline	Cash flow decreases	DSCR and score fall	Risk guidance changes
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When the revenue-decline slider moves, the dashboard recalculates DSCR, probability of default, credit score, risk grade, and covenant guidance. This allows the analyst to test a practical what-if scenario.

3. Machine learning: how risk is predicted

The project includes a Python machine-learning pipeline. It learns patterns from borrower data and produces a probability-based risk estimate for new records.

Training workflow

- Download the selected public Kaggle credit-risk dataset. It contains 5,000 synthetic borrower records.
- Remove ID fields so an identifier cannot influence the model.
- Prepare numerical data with imputation and scaling; prepare categories with one-hot encoding.
- Split data into training and holdout sets using stratified sampling.
- Train a class-balanced logistic-regression classifier and calibrate probability outputs.
- Save a reusable model artifact and evaluation metrics.
- Score a new CSV to create **probability_of_default** and **risk_decision** columns.

Example input signals

Age, employment status, annual income, monthly balance, credit score, open loans, outstanding debt, late-payment count, prior default history, fraud flag, and loan application amount.

Why logistic regression?

It is relatively simple, fast, and explainable compared with many complex models. This is useful when the model must support an analyst rather than operate as a black box.

4. Validation, limits, and responsible use

The model was evaluated using a stratified 80/20 split. It learned from 4,000 records and was evaluated on 1,000 records that were not used in training.

Metric	Result	Plain-language meaning
ROC-AUC	0.9663	Strong separation between the two risk groups in this sample dataset.
Average precision	0.9896	Strong ability to identify high-risk examples in this sample.
Brier score	0.0627	Measures the quality of probability estimates; lower is better.

Critical limitation

The Kaggle dataset is synthetic and heavily weighted toward high-risk records. These metrics are useful for demonstrating the project but must not be treated as evidence that the model is ready for real lending decisions.

What a real deployment needs

- Representative, legally sourced loan-performance data.
- Fairness, bias, calibration, stability, and drift testing.
- Secure APIs, user authentication, audit logs, and access controls.
- Human review, regulatory approval, and adverse-action reason codes.

5. How to run and use the project

Run the dashboard

```
npm start
```

Then open `http://localhost:4173` in a browser.

Install ML dependencies

```
python3 -m pip install --user -r requirements.txt
```

Download data and train

```
python3 ml/train.py --download
```

Train with the local data

```
python3 ml/train.py
```

Score a new borrower CSV

```
python3 ml/score.py borrowers.csv
```

Run tests

```
npm test
```

Project files

Path	Purpose
<code>index.html / styles.css / app.js</code>	Dashboard structure, design, and interaction logic.
<code>ml/train.py</code>	Training and model-evaluation pipeline.
<code>ml/score.py</code>	Scores new borrowers in a CSV file.
<code>MODEL_CARD.md</code>	Model method, limits, and governance requirements.
<code>outputs/</code>	Presentation assets for GitHub sharing.